

Due on Tuesday, March 7th.

If pictures are needed/relevant, please provide them with your solutions.

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1. The electric field in some region is  $\mathbf{E} = E_0 \hat{i}$ . We will consider the electric potential difference,  $V_B - V_A$ , between two points  $A$  and  $B$  on the  $x$ - $y$  plane, with coordinates  $(0, 3L)$  and  $(4L, 0)$  respectively. Ignore the  $z$ -coordinate for this problem. Here  $E_0$  and  $L$  are positive constants.

Recall:  $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$ , with the line integral proceeding along any curve starting at point  $A$  and ending at point  $B$ .

- (a) [**2 pts.**] Sketch the situation, showing the  $x$  and  $y$  axis, the two points  $A$  and  $B$ , and the direction of the electric field.
  - (b) [**3 pts.**] Consider the straight line pointing from  $A$  to  $B$ . Any element  $d\mathbf{l}$  on this line is at the same angle with the direction of the electric field. Calling this angle  $\theta$ , we obtain  $\mathbf{E} \cdot d\mathbf{l} = E_0 |d\mathbf{l}| \cos \theta$ . Calculate  $\cos \theta$  based on your drawing.
  - (c) [**6 pts.**] Calculate  $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$ , integrating along the straight line pointing from  $A$  to  $B$ . The value of  $\cos \theta$ , calculated above, might be helpful.
  - (d) [**6 pts.**] Consider now a path consisting of two segments: a line from  $A$  to point  $C$  at  $(4, 3)$ , and a line from point  $C$  to point  $B$ . Draw the situation. Only one of these ( $AC$  or  $CB$ ) segments contribute to the line integral for  $V_B - V_A$ . Which one? Calculate  $V_B - V_A$  by integrating along this segment.
2. The electric field in some region is  $\mathbf{E} = E_0 \hat{i} + 2E_0 \hat{j}$ .
    - (a) [**3 pts.**] A planar surface of area  $2\alpha$  lies in the  $x$ - $y$  plane, i.e., the normal to the surface is in the  $z$  direction. Find the magnitude of the flux of the electric field through this surface.
    - (b) [**5 pts.**] A planar surface of area  $6\alpha$  lies parallel to the  $y$ - $z$  plane. Find the magnitude of the flux of the electric field through this surface.

3. Recall that the current through a surface (e.g., through a cross-section of a current-carrying wire) is the surface integral of the current density.
- (a) **[6 pts.]** A current  $I$  is uniformly distributed over the cross-section of a wire. The cross-section is circular, with radius  $R$ . Find the current density  $J$ , as a function of  $I$  and  $R$ .
- (b) **[6 pts.]** Now consider instead the situation that the current density is larger near the outer region of the wire cross-section than at the center:  $J = kr$ , where  $r$  is the radial distance from the center of the wire. Find the current  $I$ , as a function of  $k$  and  $R$ . You will probably need to do a two-dimensional integral in polar coordinates. (In two-dimensional polar coordinates, the area element is  $dS = r dr d\theta$ .)
4. A copper conductor has  $8.5 \times 10^{28}$  electrons per cubic meter available to carry current, i.e., the carrier density is  $8.5 \times 10^{28} \text{m}^{-3}$ . The drift velocity is found to be  $3 \times 10^{-6} \text{m/s}$ .
- (a) **[2 pts.]** Look up and report the charge of an electron in SI units.
- (b) **[4 pts.]** Calculate the current density  $J$ .
5. **[7 pts.]** The current density in some region is found to be

$$\mathbf{J} = \frac{K_0 y}{\sqrt{x^2 + y^2}} \hat{i} - \frac{K_0 x}{\sqrt{x^2 + y^2}} \hat{j}$$

Use the continuity equation to show that the charge density in this region does not change with time.